

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Second Semester of M. Tech. Examination (Mechanical, AMT)****May 2018****ME786 CNC Technology & Programming****Date: 08.05.2018, Tuesday****Time: 01:30 p.m. to 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. Section I and II must be attempted in separate answer sheets.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of scientific calculator is allowed.

SECTION – I**Q-1 Answer the question below (Each carry 1 mark).****[05]**

- (i) In finish machining of an island on a casting with CNC milling machine, an end mill with 10 mm diameter is employed. The corner points of the island are represented by (0, 0), (0, 30), (50, 30) and (50, 0). By applying cutter radius right compensation, the trajectory of the cutter will be
 - (a) (-5, 0), (-5, 35), (55, 35), (55,-5), (-5,-5)
 - (b) (0,-5), (55,-5), (55, 35), (-5, 35), (-5,-5)
 - (c) (5, 5), (5, 25), (45, 25), (45, 5), (5, 5)
 - (d) (5, 5), (45, 5), (45, 25), (5, 25), (5, 5)
- (ii) In an NC machining operation, the tool has to be moved from point (5,4) to point (7,2) along a circular path with center at (5,2). The correct G and M code for this motion is
 - (a) N010 G02 X7.0 Y2.0 I0.0 J-2.0
 - (b) N010 G03 X7.0 Y2.0 I5.0 J2.0
 - (c) N010 G01 X7.0 Y2.0 I5.0 J2.0
 - (d) N010 G00 X7.0 Y2.0 I5.0 J2.0
- (iii) The tool of an NC machine has to move along a circular arc from (5, 5) to (10, 10) while performing an operation. The center of the arc is at (10, 5). Which one of the following NC tool path commands performs the above mentioned operation?
 - (a) N010 G02 X5 Y5 R5
 - (b) N010 G03 X5 Y5 R5
 - (c) N010 G01 X10 Y10 R5
 - (d) N010 G02 X10 Y10 R5
- (iv) M30 stands for:
 - (a) End of program
 - (b) End of block
 - (c) End of tape and tape rewind
 - (d) Coolant on/ off
- (v) In a CNC program block, N002 G02 G91 X40 Z40..., G02 AND G91 refer to
 - (a) Circular interpolation in counterclockwise direction and incremental dimension
 - (b) Circular interpolation in counterclockwise direction and absolute dimension
 - (c) Circular interpolation in clockwise direction and incremental dimension
 - (d) Circular interpolation in clockwise direction and absolute dimension

Q-2(a) Describe with sketch the working and construction of recirculating ball screw used in CNC machine tools stating advantages. What is preloading?

[04]

Q-2(b) Answer any three.

[12]

- Differentiate between open loop and closed loop control system.
- Machine control unit is the heart of a NC system. Justify the statement
- How are the axes of CNC machine designated? Explain with sketch stating illustration of vertical milling machine.
- Explain Geometrical statements of APT stating examples.

Q-3 Write a manual part program for part as shown in **figure 1** (All dimensions are in mm) without using cutter radius compensation. The thickness of part is 10 mm. Assume 10 mm cutter diameter.

[14]

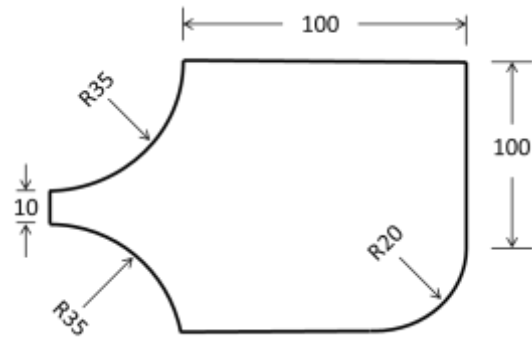


Figure 1

OR

Q-3 Write manual part program for part as shown in **figure 2** (All dimensions are in mm) with and without using canned cycle. The thickness of part is 10 mm. Assume 10 mm cutter diameter. The part origin is at 'A'.

[14]

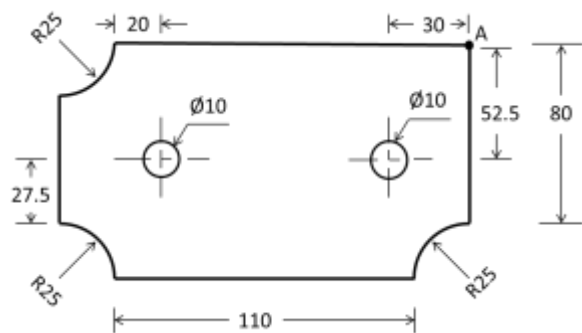


Figure 2

SECTION – II

Q-4(a) Explain cutter radius compensation cycle stating an illustration.

[03]

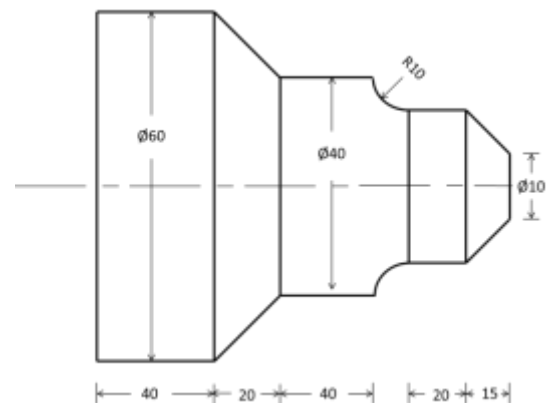
Q-4(b) What is the importance of constant surface speed in CNC turning center.

[02]

Q-5 Answer any three.**[15]**

- (i) Explain Direct Numerical Control and Distributive Numerical Control with the help of sketches. State their advantages and limitations.
- (ii) Explain advance features of CNC programming.
- (iii) Explain stating examples of turning canned cycles.
- (iv) What is the importance of using subroutines in a part program? Explain with an example.

Q-6 Write a manual part program for part shown in **figure 3** (All dimensions are in mm). Assume only finished cut is required.

[15]**Figure 3****OR**

Q-6 Write a manual part programming with and without using canned threading cycle for cutting a metric thread of 0.75 mm pitch and core diameter 12.6 mm on an aluminum rod of $\Phi 15$ mm diameter over length of 25 mm. Maximum permissible depth of cut is 0.2 mm.

G00 Rapid traverse
G01 Linear interpolation
G02 Circular interpolation CW
G03 Circular interpolation CCW
G78, G79 Mill cycle
G80 Canned cycle cancel

M00 Program stop
M02 End of program
M03 Spindle CW
M04 Spindle CCW
M05 Spindle OFF
M06 Tool change

G84 Threading cycle
G90 Absolute mode
G91 Incremental mode
G92 Presetting
M08 Coolant ON
M09 Coolant OFF
M30 End of tape
